

The Living Republic of Conscience

*Austrian Economics, Natural Genetic Engineering, and the
Design Rules of Life*

By Sage Clokey

I. The Question

What are the design rules of life?

This question has been asked by theologians, philosophers, biologists, and economists — each arriving at partial answers from within their own disciplines. But the question demands a synthesis. Life does not respect disciplinary boundaries. It is simultaneously biological, economic, moral, and spiritual. Any answer that addresses only one dimension is incomplete.

Three bodies of knowledge, when brought together, reveal a unified architecture:

Austrian economics demonstrates that complex social order arises from distributed agents acting on local knowledge, coordinated not by central command but by emergent signals — prices, profits, losses — that no single mind designs or controls.

James Shapiro's natural genetic engineering demonstrates that biological order arises the same way. Cells are not passive vehicles for randomly mutating genes. They are cognitive agents that actively restructure their own genomes in response to environmental signals, using sophisticated molecular machinery evolved for precisely that purpose.

Conscience is the principle that unifies both. It is the signal that aligns distributed agents — whether cells or citizens — toward a living order that no central authority designs or imposes.

The design rules of life are not mechanical specifications. They are the grammar of a living republic.

II. The Genome as Adapting Law

The conventional understanding of the genome treats it as a blueprint — a fixed set of instructions executed by molecular machinery. James Shapiro dismantled this metaphor. The genome is not a blueprint. It is a living, adapting law.

Every cell carries its own copy of this law. Every cell reads it, interprets it, and — under the right conditions — revises it. The mechanisms are real and well-documented: transposition, recombination, horizontal gene transfer, targeted mutagenesis under stress, epigenetic modification, and the recruitment of mobile

genetic elements to generate novel regulatory architectures. These are not errors in copying. They are acts of authorship.

Each cell possesses distributed knowledge that exists in no other cell. Its position in the tissue, its metabolic state, its history of environmental encounters, its epigenetic configuration — all of this constitutes local knowledge that cannot be centralized. No master gene, no regulatory hub, no nucleus of command possesses the totality of what the organism knows. The knowledge is distributed across billions of agents, each acting on what it alone can sense.

This is the biological knowledge problem. It is identical in structure to the economic knowledge problem that Friedrich Hayek identified in 1945: the knowledge required for rational coordination is dispersed among millions of individuals, exists in forms that cannot be articulated or transmitted to a central authority, and changes constantly. No central planner can possess it. Therefore, no central planner can substitute for the distributed process that uses it.

In biology, the cell is the citizen. Its genome is its personal law — the rules it follows, the capacities it carries, the constraints it respects. But this law is not frozen. It adapts to the cell and the cell adapts to its environment. And the cell does not act alone. It exists in a network of cells adapting together, verifying each other's work through molecular signaling, immune surveillance, and apoptotic quality control — a biological analog to distributed verification. Every change that works persists. Every change that fails is pruned. The soil of successful revisions accumulates, and from that soil mountains grow.

This is not random drift. It is distributed, directed trial and error — the same process by which markets discover prices, entrepreneurs discover opportunities, and common law discovers justice.

III. Concept of a Plan

The distinction between a **concept of a plan** and a **central plan** is the hinge on which everything turns.

A central plan is a rigid blueprint. It is designed in advance by a central authority, imposed on all participants, and does not change in response to the free choices of those who must live under it. It assumes that the planner possesses sufficient knowledge to dictate outcomes. It does not account for free will because free will is an obstacle to its execution. The central plan treats the world as a machine to be programmed.

A concept of a plan is alive. It is a living order that adapts to the free will of the world. It sets the trajectory — the deep grammar, the moral direction, the arc of creation — but it does not dictate the specific steps. It responds to the choices of those involved. It bends without breaking. It absorbs surprise, incorporates discovery, and grows richer through the participation of free agents. The concept of a plan is the plan of spontaneous order.

God's plan, understood this way, is not a central plan. It is a concept of a plan. It is not imposed upon creation. It unfolds through the voluntary actions of living beings. It preserves free will because free will is not an obstacle to it — free will is the mechanism through which it operates.

This is precisely what Hayek meant by spontaneous order, extended to its theological ground. The market order is not designed by any participant, yet it exhibits coherence, adaptability, and a capacity for self-correction that no designed system can match. It works

because millions of agents act freely on local knowledge, coordinated by signals — prices — that emerge from their interactions. No one plans the market. Yet the market has a plan — not a central plan, but a concept of a plan that adapts to every choice made within it.

The genome works the same way. No central gene plans the organism. Yet the organism exhibits breathtaking coherence — trillions of cells coordinated across dozens of tissue types, maintaining homeostasis, repairing damage, adapting to stress, and reproducing with fidelity. This coherence is not imposed from above. It emerges from below, through distributed agents following local rules and revising those rules in response to reality.

Austrian economics proved that central economic planning fails because it cannot solve the knowledge problem. Shapiro proved that central genetic planning does not exist because the genome is a read-write system operated by distributed cellular agents. The lesson is the same in both cases: **living order cannot be centrally commanded. It must emerge from the free action of distributed agents aligned with a concept of a plan.**

IV. Conscience as the Coordinating Signal

If order is not centrally imposed, what coordinates it?

In economics, the answer is the price system — an emergent signaling network that communicates dispersed knowledge without any participant needing to understand the whole.

In biology, the answer is molecular signaling — hormones, cytokines, neurotransmitters, quorum-sensing molecules, and the countless other signals by which cells communicate local conditions and coordinate collective responses.

But these are mechanisms, not sources. They describe *how* coordination occurs, not *why* it tends toward coherence rather than chaos. Prices coordinate economic action, but what ensures that the market order tends toward productivity rather than destruction? Molecular signals coordinate cellular action, but what ensures that the organism tends toward life rather than cancer?

The answer is conscience.

Conscience is a property of living systems. It is the invisible hand — not of the market alone, but of creation itself. It is the signal that orients distributed agents toward harmony with the living order. In human beings, conscience is experienced directly: the inner knowledge of right and wrong that precedes any written law, any cultural tradition, any institutional authority. In biological systems, conscience operates through the deep grammar of life — the structural and functional constraints that channel genetic experimentation toward viability, cooperation, and increasing complexity.

Conscience does not compel. It guides. It preserves free will absolutely. A cell can ignore its signaling environment and proliferate without restraint — this is cancer. A person can ignore conscience and act against what they know to be right — this is sin. In both cases, the system provides correction: immune surveillance eliminates cancerous cells; the consequences of wrongdoing impose their own discipline. But the freedom to choose wrongly is never revoked. It is essential to the architecture.

This is what distinguishes a living order from a mechanical one. A machine cannot choose wrongly because it cannot choose at all. A living being can choose wrongly because it is genuinely free. The coherence of life is therefore not the coherence of a machine but the coherence of a republic — achieved through countless free acts tending toward alignment with a higher order.

V. The Living Republic

America was not founded as the nation of one religion, one church, or one sect. It was founded on the principle that every person possesses a conscience and bears responsibility for how they use it.

Faith is not ultimately found in belonging to a particular denomination, memorizing a particular text, or accepting a particular historical tradition. Those things can guide, but faith is revealed in the relationship between each person and their Creator — a relationship expressed by freely choosing to do what is right.

Our souls are entrusted to us alone. They are not the responsibility of kings, priests, popes, rabbis, gurus, sages, politicians, or any earthly authority. Each person must answer for their own actions. "I was only following orders" is never enough when those orders violate conscience.

Every written law is humanity's attempt to approximate the higher law already written on the human heart. Written law is imperfect because human beings are imperfect. Therefore, no written code should ever be followed blindly. Conscience must always remain awake.

This is the biological principle made civic. Just as the genome is a written law that must be continually revised to remain viable, human law is a written code that must be continually revised to remain just. And just as the cell revises its genome through distributed experimentation — not by waiting for orders from a master gene — citizens must revise their laws through distributed moral action, not by waiting for commands from a ruling class.

There is no chain of command between human beings. All are equal and independent. This is not merely a political ideal — it is a structural requirement. The wisdom of crowds depends on independence. When agents imitate each other or defer to a central authority, the distributed knowledge that makes collective intelligence possible is destroyed. Each person must act independently from every other but from the same source — conscience, the higher law, the concept of a plan written on every human heart.

No mere human should be the intermediary of this relationship. If a man's command contradicts God's command, do God's command. If a man's command accords with God's command, do God's command. The source is always the same. The authority is always the same. Human intermediaries may persuade, but they may never compel. Persuasion respects the independence of the agent. The initiation of force destroys it.

Every person has the obligation not merely to obey leaders but to keep leaders on the right path by refusing to participate in wrongdoing. As Fives declared, "I cannot just follow orders when I know they're wrong." That principle reflects a truth older than any government: there is a higher order than the written order.

It is everyone's obligation to do what is right in whatever they are doing. It should always be the first priority — before power, before money, before fame, before comfort, before the approval of the crowd. Mob rule is not inherently moral. The majority is not always right. In fact, the majority is wrong more often than it admits, and the ruling class is wrong even more often than the majority. But this does not mean that truth is unknowable. It means that truth must be sought by each person independently, through conscience, not received passively from any human authority.

This is the Sagent of the crowds. Not the mere wisdom of crowds — which can degenerate into mob rule when independence is lost — but the *sage* intelligence that emerges when each agent acts from conscience, independently, aligned with the same source. The word carries its meaning: sage as in wise, agent as in one who acts. A Sagent is one who acts wisely from conscience.

Protection itself must be measured by this standard. Those who claim to protect but initiate force are not protectors — they are aggressors wearing the language of protection. You cannot protect by first initiating violence. Defense is the answer to aggression, not its mirror. The obligation to protect is the obligation to guard against *all* initiations of force, including — especially — those that come wrapped in the authority of the state, the church, the corporation, or the crowd.

VI. The Biological Proof

Biology provides the empirical demonstration of these principles.

Every living organism is a republic of cells. Each cell carries its own adapting law — its genome. Each cell possesses local knowledge that exists nowhere else in the organism. Each cell acts on that knowledge, responding to its environment, revising its gene expression, and — when conditions demand — restructuring its genome itself.

No cell commands all others. The nucleus does not command the cell; it is a library the cell consults. The brain does not command the body; it is a coordination center that integrates signals from billions of distributed agents. Hierarchy exists in biology, but it is hierarchy of function, not hierarchy of command. The heart does not obey the brain out of submission. It beats because its own cellular law tells it to beat. Coordination emerges because each cell follows its own law while remaining responsive to the signals of its neighbors.

When this distributed republic functions well, the result is health — an organism of staggering complexity operating in dynamic harmony with its environment. When it fails — when a cell stops listening to its neighbors, stops responding to signals, stops revising its law in response to reality — the result is cancer: unchecked growth, resource hoarding, destruction of the surrounding community. Cancer is biological tyranny. It is the cell that has abandoned conscience.

James Shapiro's work reveals that this republic is not merely maintaining a fixed order. It is *creating* new order. Natural genetic engineering is the process by which cells generate novel genomic architectures — new regulatory networks, new protein functions, new developmental programs. This creativity is not random. It is channeled by the deep grammar of molecular biology: the structural constraints of protein folding, the logic of gene regulation, the

requirements of cellular cooperation. The grammar does not dictate what is written. It defines what can be written coherently.

This is where artificial intelligence enters — not as a replacement for the living process but as an instrument for reading its grammar.

VII. AI as the Reader of Grammar

Protein language models like ESM-3, genomic foundation models, and structural prediction tools like AlphaFold are trained on the accumulated text of billions of years of distributed genetic experimentation. Every protein sequence in every database is a sentence written by natural genetic engineering — tested by reality, revised by selection, and preserved because it works.

What these models learn is not the plan. They learn the grammar — the deep structural rules by which living systems compose functional proteins, regulatory networks, and genomic architectures. They learn which amino acid sequences fold into stable structures. They learn which regulatory motifs activate gene expression in response to specific signals. They learn the patterns of co-evolution that reveal functional relationships between proteins.

This grammar is precisely what Austrian economics would predict: it is discoverable but not designable from first principles. You cannot sit in an office and deduce the rules of protein folding from pure reason, just as you cannot deduce the correct price of wheat from pure reason. You must learn from the data of actual living exchange — from the billions of experiments that cells have already conducted and the results that reality has already verified.

The English-to-Protein bridge — the capacity to describe a desired biological function in natural language and have AI translate that description into a viable protein sequence — is therefore not an act of central planning. It is an act of literacy. It is learning to speak the language that life already speaks. The grammar was not invented by the model. It was discovered in the accumulated wisdom of living systems.

The danger, as always, is the temptation to treat this capacity as central planning power — to imagine that because we can read the grammar, we can dictate the text. This is the biological equivalent of the fatal conceit that Hayek identified in economics: the belief that because we can observe the market order, we can design a better one from scratch. We cannot. The order is too complex, too dependent on distributed knowledge, too alive to be replaced by any human design.

The proper use of these tools is not to command life but to converse with it — to propose genetic revisions that are grammatically coherent with life's living order and let reality verify them through the same distributed process that has always governed biological creation.

VIII. The Design Rules

The design rules of life, synthesized from Austrian economics, natural genetic engineering, and the principle of conscience, are these:

Subsidiarity. Decisions are made at the lowest competent level. The cell decides what the cell can decide. The tissue decides what the tissue can decide. The organism decides what the organism can decide. No level usurps the authority of another. This is the knowledge problem applied universally: centralization destroys the distributed knowledge that makes intelligent coordination possible.

Revisability. All written law — genetic, civic, or moral — must remain open to revision. Frozen systems die. The genome is a read-write system. The Constitution has an amendment process. Common law evolves through cases. Dogma that cannot be questioned is a genome that cannot adapt. It is a death sentence delivered in the language of permanence.

Independence. Each agent must act from its own knowledge and its own conscience. Imitation, conformity, and deference to authority destroy the independence that makes distributed intelligence possible. The Sagent of the crowds requires that each agent think independently from the same source — not from each other, and not from any human intermediary.

Coherence through conscience, not coercion. Unity is achieved through countless free acts tending toward alignment with the higher law, not through commands imposed from above. Persuasion is legitimate. The initiation of force is not. A living order that depends on coercion is not a living order — it is a machine wearing the mask of life.

Stress as signal. Environmental stress activates natural genetic engineering. Economic stress — losses, failures, recessions — signals malinvestment and triggers correction. Moral stress — guilt, shame, the quiet voice that says *this is wrong* — signals deviation from conscience. In all three domains, stress is not a failure of the

system. It is the system working. It is the call to revise the written law.

Distributed verification. No revision is accepted on the authority of the reviser alone. In biology, every genetic change is tested by the environment and verified by the network of cells that must live with it. In economics, every entrepreneurial venture is tested by the market. In morality, every action is tested by its consequences and by the conscience of others who witness it. Authority is earned through verification, not claimed through position.

Non-aggression. Protection means defense against the initiation of force, not the initiation of force in the name of protection. This applies to cells: the immune system destroys invaders and cancerous cells but does not attack healthy tissue. It applies to citizens: the obligation to protect includes the obligation to guard against those who claim protective authority while initiating violence. You cannot grow a living order through coercion. You can only grow it through free action aligned with conscience.

IX. The Kingdom of Conscience

The Kingdom of Heaven is not a centralized hierarchy. It is a living republic of conscience.

Every living being acts as a distributed agent, participating in the continual unfolding of God's concept of a plan. Every cell revises its genome. Every person revises their understanding of right and wrong. Every generation revises its written laws. The process never ends because the concept of a plan is not a destination — it is a

direction. It is the living arc of creation bending toward justice, toward truth, toward harmony — not because it is forced to bend, but because billions of free agents choose, in their imperfect and stumbling way, to align their actions with what conscience tells them is right.

The highest form of faith is not the repetition of words but the courage to choose what is right.

The highest form of order is not the obedience of subjects but the alignment of free agents.

The highest form of law is not the written code but the living conscience that the written code attempts to approximate.

This is the Living Republic. This is the Living Age.

The creation of the Living Age requires media to explain why and technology to show how. This essay is the why. The genome design system — the English-to-Protein bridge, the compatibility engine, the adaptive automation — is the how. Both are expressions of the same truth: that life is not a machine to be programmed but a republic to be cultivated.